

PMBOK / PMI | PMB-17

Quality Management Plan

How quality requirements will be met and demonstrated

Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

Field	Detail
Document title	Quality Management Plan
Document reference	PMB-17-HEL-T2-v1.0
Project / Programme	Helios Tranche 2 — Scale Build
Version	1.0
Status	Baselined
Date	18 Sep 2026
Author	D. Okonkwo, Programme Manager
Owner	Project manager with quality specialists
Approver	Sponsor
Classification	Internal
Retention	7 years from programme closure

Document control

Version history

Version	Date	Author	Summary of change
0.1	04 Jun 2026	D. Okonkwo, Programme Manager	Initial draft
0.2	21 Jul 2026	D. Okonkwo, Programme Manager	Updated following workstream lead review
1.0	18 Sep 2026	D. Okonkwo, Programme Manager	Baselined at the Tranche 2 funding gate

Reviewers

Name	Role	Review scope	Date reviewed
T. Nakamura, Chief Platform Architect	Chief Platform Architect	Technical content and solution alignment	08 Sep 2026
L. Haddad, CISO	CISO	Security, assurance and residency content	10 Sep 2026
C. Whitfield, Finance Business Partner	Finance Business Partner	Cost, funding and benefit figures	11 Sep 2026
M. Verity, Head of P3O	Head of P3O	Governance standards and completeness	15 Sep 2026

Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

Name	Role	Decision	Date
R. Al Mazrouei, Group CTO	Senior Responsible Owner / Group CTO	Approved	18 Sep 2026
H. Farrell, Director of Infrastructure	Director of Infrastructure	Approved	18 Sep 2026
L. Haddad, CISO	CISO	Approved subject to closure of ISS-003	18 Sep 2026

Related products

Reference	Product	Relationship
PMB-08	Requirements Documentation and Traceability Matrix	Source of acceptance criteria
PMB-06	Project Management Plan	Parent plan
P2-08	Quality Management Approach	PRINCE2 equivalent

How to use this template

Aspect	Guidance
Purpose	Describes how the project will demonstrate compliance with quality policies, standards and requirements — the quality standards applying, the assurance and control activities, the metrics and the roles responsible.
Framework	PMBOK / PMI — Quality Management — Plan Quality Management
Created when	During planning
Reviewed / updated	At each phase gate
Owner	Project manager with quality specialists
Approver	Sponsor
Format options	Document, presentation, register entry, or tool record — the content matters, not the medium.
Tailoring	Distinguish assurance (are we following the process) from control (is the product right). Conflating them means one of the two does not happen.

This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.

Quality criteria

Use these to check the product is fit for purpose before submitting it for approval.

- Quality standards are named specifically, with versions
- Assurance and control activities are distinguished
- Metrics have thresholds and a defined response when breached
- Independence of assurance from delivery is addressed
- The cost of quality is considered — both conformance and non-conformance

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1. Quality standards applying

Standard	Version	Applies to	Mandatory?	Owner
Any deviation requires a recorded waiver accepted by the CISO with a stated expiry	v1.0	Facilities, power and cooling works within the Tranche 2 boundary	Yes	R. Al Mazrouei, Group CTO
Enforced automatically in the pipeline; manual assertion is not accepted as evidence	v1.0	DXB-1 halls A and B, 2,048 accelerators and the supporting fabric and storage	Partially	P. Raghavan, Platform Engineering Lead
Reviewed at each end-of-tranche review and updated by change control only	v1.0	The three onboarding tenants and their committed migration workloads	Yes	V. Chandra, Product Management
Delegated to the Programme Manager within tolerance; above tolerance to the Programme Board	v1.0	The three onboarding tenants and their committed migration workloads	No	M. Verity, Head of P3O

2. Quality objectives and metrics

Ref	Objective	Metric	Target	Threshold	Measured by	Frequency
QOA-01	Provide evidence that platform controls operate as designed	Open critical security findings	AED 5.1m per year	Coolant leak alerts the NOC within 60 seconds and isolates the affected rack	R. Al Mazrouei, Group CTO	Monthly
QOA-02	Provide evidence that platform controls operate as designed	All-reduce efficiency at 256-GPU scale	Under 2 hours	Storage sustains 1.2 TB/s against a representative training corpus for 30 minutes	A. Kowalski, Data Centre Lead	Quarterly
QOA-03	Confirm that Tranche 2 remains viable, achievable and within tolerance	Rolling 12-month PUE	Zero open critical findings	Median environment request-to-usable time under two hours across 20 sampled requests	D. Okonkwo, Release Train Engineer	Fortnightly
QOA	Track benefit	GPU utilisation,	92% at	All-reduce	L. Haddad,	Each

Ref	Objective	Metric	Target	Threshold	Measured by	Frequency
-04	realisation from Tranche 2 closure through to FY29	rolling 30-day average	256-GPU job scale	efficiency at or above 92% on three consecutive 256-GPU runs	CISO	planning interval

3. Quality assurance activities

Process-focused. Are we doing the right things in the right way?

Activity	Scope	Performed by	Independent of delivery?	Frequency	Output
Complete hall B structural survey	Facilities, power and cooling works within the Tranche 2 boundary	I. Petrova, Data Platform Lead	Yes	Monthly	Parallel filesystem accepted at 1.2 TB/s sustained
Place structural strengthening order	Platform, MLOps and security workstreams across the Agile Release Train	J. Ferreira, Network Lead	No	Quarterly	400G fabric commissioned and NCCL-validated
Execute floor strengthening works	Excludes AUH-2, model development and business-unit data engineering	V. Chandra, Head of AI Enablement	Yes	Fortnightly	Platform v1.0 — scheduling, quota, observability and image supply chain
Lodge utility uplift application	The three onboarding tenants and their committed migration workloads	S. Bhatt, Project Manager	Partially	Each planning interval	Hall A accepted at 1,024 GPUs with acceptance certificate
Install switchgear and energise 6 MVA supply	DXB-1 halls A and B, 2,048 accelerators and the supporting fabric and storage	C. Whitfield, Finance Business Partner	Yes	Weekly	Independent security assurance report with findings closed
Install CDUs and secondary cooling loop	Facilities, power and cooling works within the Tranche 2 boundary	R. Al Mazrouei, Group CTO	No	Each iteration	Operations runbook set with two completed live drills

4. Quality control activities

Product-focused. Is the deliverable right?

Deliverable	Control method	Criteria	Performed by	Records retained
Hall A power and cooling installation	e.g. Peer review / Test / Inspection / Penetration test	Sustains 60 kW per rack for 4 hours with return temperature within design	A. Kowalski, Data Centre Lead, M&E contractor, independent commissioning agent	Retained in the facilities evidence pack, 7 years
400G fabric — hall A	NCCL all-reduce benchmark at 256-GPU scale	At or above 92% efficiency on three consecutive runs	J. Ferreira, Network Lead, vendor performance engineering	Retained in the network evidence pack, 7 years
Parallel filesystem tier	Workload-representative throughput benchmark	1.2 TB/s sustained for 30 minutes against the fraud analytics corpus	I. Petrova, Data Platform Lead, storage vendor	Retained in the storage evidence pack, 7 years
Accelerator batch — hall A racks 1 to 16	Firmware hash verification against the vendor SBOM, then 72-hour burn-in	All hashes match; no thermal or ECC faults during burn-in	K. Adeyemi, Security Architect, infrastructure engineers	Retained in the security evidence pack, 7 years
Platform v1.0 — scheduling and quota	72-hour contention test across three tenants	Fair-share quota holds within 5% for every tenant	P. Raghavan, Platform Engineering Lead, tenant representatives	Retained in the platform evidence pack, 7 years
Image supply chain and admission control	Negative test — attempt admission of unsigned and tampered images	All attempts refused and alerted within 30 seconds	K. Adeyemi, Security Architect, Cordant Assurance	Retained in the security evidence pack, 7 years
Multi-tenant platform — security posture	Independent penetration test	No open critical or high findings at retest	Cordant Assurance, CISO	Retained in the assurance evidence pack, 7 years
Operations runbook set and drills	Two witnessed live coolant leak drills	NOC alerted within 60 seconds; rack isolated; runbook followed without deviation	G. Lindqvist, Service Operations Lead, FM provider	Retained in the operations evidence pack, 7 years

5. Roles and responsibilities

Role	Quality responsibility	Reports to
CISO and Security Architect	Fair-share quota holds within 5% for each tenant under contention	R. Al Mazrouei, Group CTO

Role	Quality responsibility	Reports to
Business Owners and tenant representatives	Storage sustains 1.2 TB/s against a representative training corpus for 30 minutes	P. Raghavan, Platform Engineering Lead
Service Operations and FM provider	All-reduce efficiency at or above 92% on three consecutive 256-GPU runs	V. Chandra, Product Management
Agile Release Train — all teams	Storage sustains 1.2 TB/s against a representative training corpus for 30 minutes	M. Verity, Head of P3O

6. Non-conformance handling

Severity	Definition	Response	Authority to accept
Critical	Programme objectives cannot be met without resolution — response within four hours, escalated to the SRO immediately	Stop and remediate	D. Okonkwo, Programme Manager
Major	Exceeds tolerance — Programme Board decision at the next monthly meeting	Exceeds tolerance — Programme Board decision at the next monthly meeting	D. Okonkwo, Programme Manager
Minor	No baseline impact — decided by the Programme Manager within 2 working days	No baseline impact — decided by the Programme Manager within 2 working days	D. Okonkwo, Programme Manager

7. Cost of quality

Category	Activities	Estimated cost
Prevention	Design review at each freeze point, coding standards, automated policy checks in the pipeline — AED 1,100,000	AED 4,200,000
Appraisal	Penetration testing, acceptance benchmarking, independent assurance review — AED 1,900,000	AED 4,200,000
Internal failure	Rework identified before handover, principally fabric retuning and burn-in reruns — AED 620,000	AED 4,200,000
External failure	Defects reaching production during warranty, budgeted at AED 340,000	AED 4,200,000

8. Continuous improvement

How quality findings feed back into process improvement during the project rather than only at closure.

Three themes have emerged from Tranche 1: security assurance run only at completion finds expensive problems late; specialist vendor performance engineering is worth budgeting as a planned activity rather than a remedial one; and stakeholder attitude assessed from a single early conversation is unreliable where autonomy is being removed.